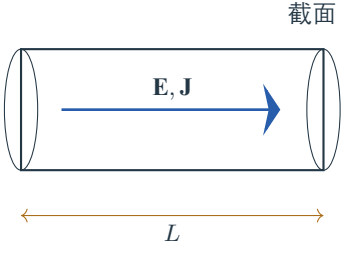
从局域欧姆定律到均匀导体的端口关系



$$U = EL, \quad I = jS$$

$$j = \sigma E \quad \Rightarrow \quad U = IL/(\sigma S)$$

$$R = L/(\sigma S) = \rho L/S$$

忽略边缘效应，电场和电流密度沿轴向均匀